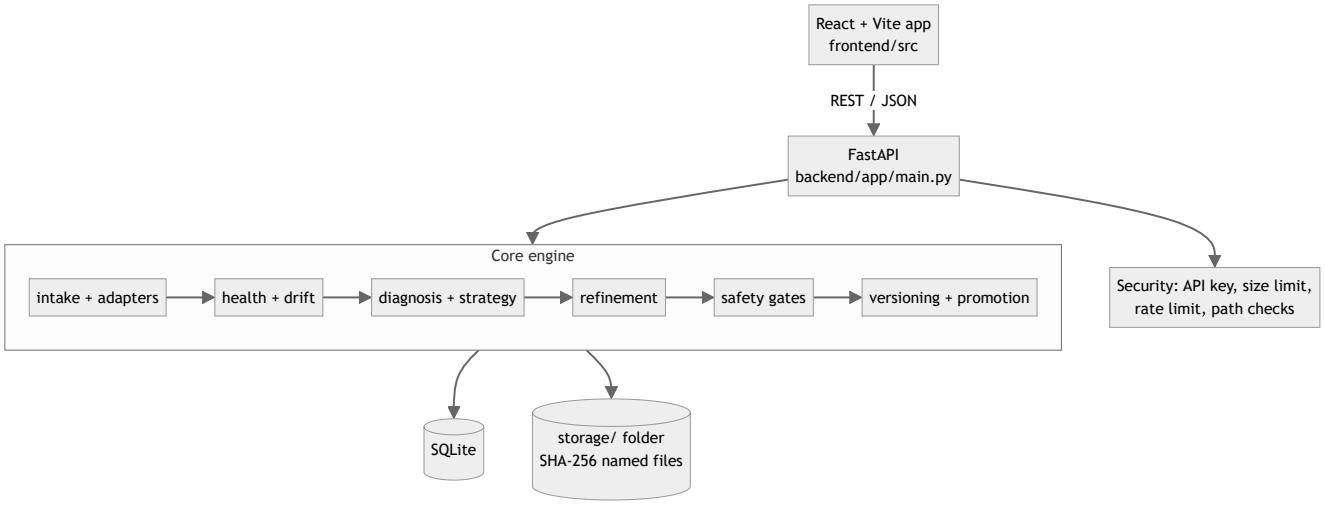
AI MODEL HEALTH MONITORING & AUTOMATED MAINTENANCE SYSTEM

Technical Requirements Document

Draft · 2026-09-29

1. Architecture



Modular monolith: one backend process, one database file, one storage folder.

2. Modules

Area	Files	Job
API	`main.py`	~50 routes, error handlers
Storage	`artifact_storage.py`, `intake.py`, `sandbox.py`	Hash, save, first-load check
Models	`adapters/*`	One adapter per framework, chosen by `detector.py`
Tasks	`tasks/*`	Metrics per task type
Health	`health.py`, `health_status.py`, `drift.py`, `prediction_drift.py`, `data_quality.py`, `schema_checker.py`	5-pillar score
Fixing	`diagnosis.py`, `strategy_selector.py`, `refinement.py`, `retraining.py`	Cause, then targeted fix
Safety	`safety_gates.py`, `promotion.py`, `versioning.py`	Gates, promote, rollback
Data	`database.py`	Tables and queries

3. Main data

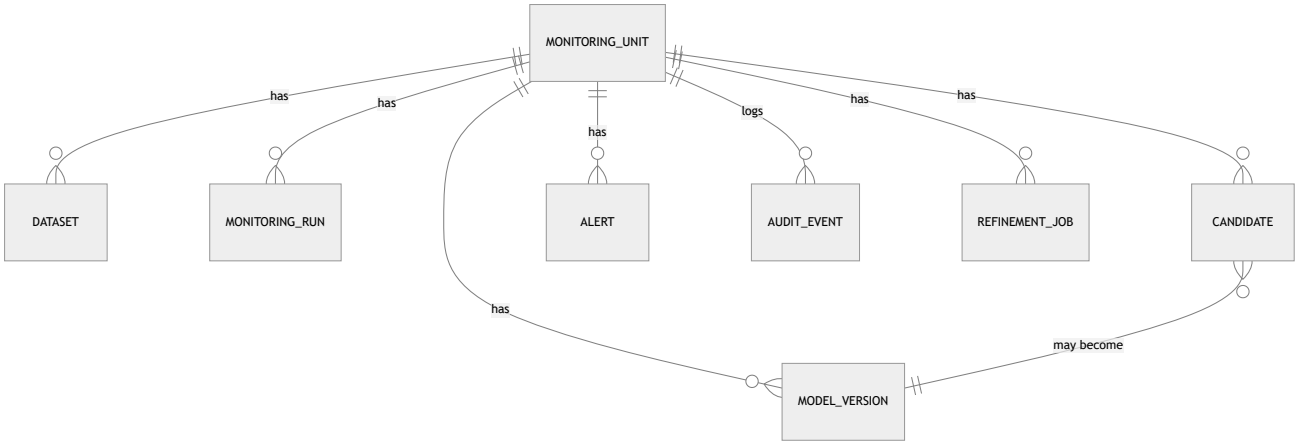


Table names are indicative; check `database.py` for exact columns.

4. Key technical rules

- **Drift:** KS-test (numeric) and TVD (categorical); no external drift library.
- **Diagnosis to fix:** Unexpected drift → data refresh. Decay → bounded random search. Imbalance → class balancing. Missing values → imputation. Constant columns → pruning. Output shift → calibration.
- **7 gates:** integrity, reload, compatibility, dry-run, absolute floor, improvement ≥ +0.005, no secondary metric drop worse than −0.02.
- **Modes:** AUTO promotes; ASSISTED stages; MANUAL advises.
- **Storage:** file name = SHA-256, so duplicates are stored once.
- **Errors:** custom exceptions; no stack traces to clients.

5. Non-functional targets

Topic	Target
Upload limit	50 MB (`MAX_UPLOAD_SIZE_MB`)
Rate limit	Intake 30/min, refine 10/min per IP
First-load timeout	30 s
Isolation	Failure in one unit must not change another
Tests	27 scripts pass; move to `pytest` with a separate test DB

6. Config

`MODEL_HEALTH_API_KEY`, `MAX_UPLOAD_SIZE_MB`, `ALLOWED_ORIGINS`.

7. Technical debt

1. Tests share the dev DB and are not pytest.
2. In-memory rate limiter breaks with more than one server process.
3. Refinement runs inside the API process (no job queue).

4. SQLite + local disk = single server.
5. `ARCHITECTURE.md` is long and partly out of date on uncommitted work.